Q1/

$f(x) = 5 - 2x$ → polynomial of degree 1 @ linear function

$g(x) = \frac{2}{x+2}$ → rational function

$$\frac{f(-2) - g(5)}{3} = \frac{(5 - (-4)) - \frac{2}{7}}{3}$$

$$= \frac{61}{21} \#$$

Q2

$$\frac{2f(-1)+g(-3)}{2} = \frac{2\left[\frac{-1+3}{1-2(-1)}\right]+\frac{1}{-3+5}}{2}$$

$$= \frac{\frac{4}{3}+\frac{1}{2}}{2}$$

$$= \frac{11}{12} \text{ Ans}$$

Q3

		Set Notation	Interval Notation
(a) $1-2x$	polynomial D: all real numbers	$x \in \mathbb{R}$	$(-\infty, +\infty)$
(b) $\frac{x}{1-2x}$	$1-2x \neq 0$ $2x \neq 1$	$x \neq \frac{1}{2}$	$(-\infty, \frac{1}{2}), (\frac{1}{2}, +\infty)$
(c) $\frac{x}{\sqrt{x+1}}$	$x+1 > 0$	$x > -1$	$(-1, +\infty)$

Q4/ (a) $f(x) = \dfrac{36}{a+bx}$

$f(9) = \dfrac{36^{3}}{a+9b} = 12 \longrightarrow 3 = a+9b$ — ①

$f(8) = \dfrac{36^{4}}{a+8b} = 9 \longrightarrow 4 = a+8b$ — ②

① − ②: $b = -1$ #

From ①: $a = 3 - 9(-1) = 12$ #

(b) $f(x) = 4$

$$\frac{36}{12-x} = 4$$

$$36 = 48 - 4x$$

$$4x = 12$$

$$x = 3$$ #

(c) $f(x) = x$

$$\frac{36}{12-x} = x$$

$$36 = 12x - x^2$$

$$x^2 - 12x + 36 = 0$$

$$(x-6)(x-6) = 0$$

$$x = 6$$ #

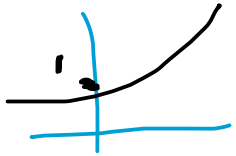
Q5/

(a) $e^{2x} - 3e^x = 0$

$(e^x)^2 - 3e^x = 0$

$e^x(e^x - 3) = 0$

$e^x = 0$



not possible

$e^x = 3$

$x = \ln 3$ #

(b) $(e^x)^2 - 7e^x + 12 = 0$

Let $p = e^x$;

$p^2 - 7p + 12 = 0$

$(p-3)(p-4) = 0$

$e^x = 3, 4$

$\therefore x = \ln 3, \ln 4$ #

(c) $3e^x - 5 = -2$

$\leftarrow \overline{e^x}$

$3(e^x)^2 - 5e^x + 2 = 0$

$(3e^x - 2)(e^x - 1) = 0$

$3e^x - 2 = 0$

$x = \ln\left(\frac{2}{3}\right)$ //

$e^x - 1 = 0$

$x = \ln 1$

$= 0$ //

Q6

(a) $\log_3\left(\frac{1}{9}\right) = x$

$3^x = \frac{1}{9}$

$= \frac{1}{3^2}$

$= 3^{-2}$

$\therefore x = -2$ //

(b)

$$5^{x+3} = 100$$

$$(x+3)\lg 5 = \lg 100$$

$$x+3 = \frac{\lg 100}{\lg 5}$$

$$= \ldots\ldots$$

$$= \ldots\ldots - 3$$

$$\therefore x = -0.139 \text{ (3 sf)}$$

(c)

$$\log_3 x^2 = \log_3 3 + \log_3 (x+6)$$

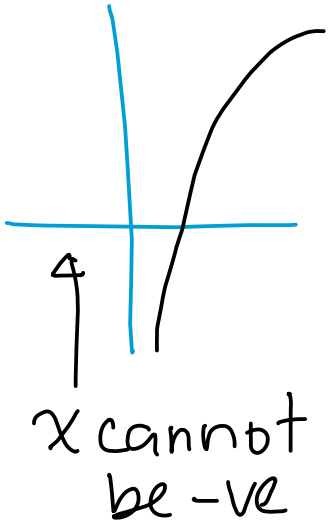
$$= \log_3 3(x+6)$$

$$x^2 - 3x - 18 = 0$$

$$(x+3)(x-6) = 0$$

$x = -3$ (reject)

$x = 6$



(d)

$$\log_x 6^5 - \log_x 96 = 4$$

$$\log_x \frac{6^5}{96} = 4$$

$$x^4 = 81$$

$$= 3^4$$

$$\therefore x = 3 \#$$

(e)

$$\lg 7^{x+1} = \lg 2^{3x-2}$$

$$(x+1)\lg 7 = (3x-2)\lg 2$$

$$x\lg 7 + \lg 7 = 3x\lg 2 - 2\lg 2$$

$$x(\lg 7 - 3\lg 2) = -2\lg 2 - \lg 7$$

$$x = \frac{-2\lg 2 - \lg 7}{\lg 7 - 3\lg 2}$$

$$= 24.954 \#$$